DS 3010 Homework 6 Solutions

Problem 1: Concept Review

(a) Best subset selection will produce a collection of p models $M_1, M_2, \ldots, M_p$. These represent the 'best' model of each size (where 'best' here is defined as the model with the smallest RSS).

i **True or False**

False. It is not true that the model identified as M_{k+1} must contain a subset of the predictors found in M_k using the subset selection algorithm.

ii Explain your answer.

Since subset selection is an exhaustive search, it will consider the best model for each size $k+1$, regardless of what was chosen as the best model for size k.

(b) Same question as part (a) but instead of best subset selection, we now carry out forward selection.

Yes, by design forward selection only considers moving forward with predictors in the previous model. So M_{k+1} must contain a subset of predictors found in M_k.

(c) Suppose we perform best subset selection, forward selection, and backward selection on a training data set.

Best subset selection produces a model with k predictors for each $k = 1, \ldots, p$. Denote this model by $M_{k,\textbf{subset}}$. For example, when $k = 5$, $M_{5,\text{subset}}$ is the best 5-predictor model according to best subset selection.

Similarly, forward selection produces a model with k predictors, denoted by $M_{k,\textbf{forward}}$, and backward selection produces a model with k predictors, denoted by $M_{k,\textbf{backward}}$.

For each case below, state which of the three models would be expected to have the smallest *training* MSE, and briefly explain why.

(i) $k = 1$

For $k = 1$, subset has the same model as forward stepwise and therefore the same RSS (and training MSE). Backward stepwise may have a different model, and a worse RSS.

(ii) $k \in \{2, ..., p-2\}$

For $k \in \{2, ..., p-2\}$, subset has equal or better training MSE than forward stepwise and backward stepwise.

(iii) $k = p - 1$

For $k = p - 1$, subset has the same model as backward stepwise and therefore the same RSS (and training MSE). Forward stepwise may have a different model, and a worse RSS.

(iv) $k = p$

For $k = p$, subset, forward, and backward have the same model and therefore the same RSS (and training MSE).

(v) Based on your answers above, generalize your conclusion for any $k = 1, \ldots, p$.

In summary, for $k = 1, \ldots, p$ the subset model has equal or better training MSE than the other methods. Therefore, we expect $M_{k,subset}$ to give the model with the smallest training MSE.

(d) Same setup as part (c). For a given k, which of these three models has the smallest test MSE? Explain your answer.

(i) $k = 1$

Subset and forward stepwise have the same model when $k = 1$, so they have the same test MSE. Backward selection may produce a different model, so its test MSE may differ.

(ii) $k = p$

For $k = p$, all three methods are the same and hence have the same test MSE.

(iii) $k = p - 1$

Subset and backward have the same model when $k = p - 1$, so they have the same test MSE. Forward selection may produce a different model, so its test MSE may differ.

Other than the cases where the fitted models are identical (e.g., $k = 1$, $k = p$, $k = p - 1$ as described above), it is not possible to guarantee which method yields the smallest test MSE. Test MSE depends on generalization and can vary due to randomness in the data and the train/test split.

Problem 2: Simulation Studies

We have seen that as the number of features used in a model increases, the training error will necessarily decrease, but the test error may not. We will now explore this in a simulated data set. Start by generating a data set with $p = 20$ features, $n = 1000$ observations, and an associated response vector Y generated according to the model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots \beta_p X_p + \epsilon,$$

You can use this code to get started:

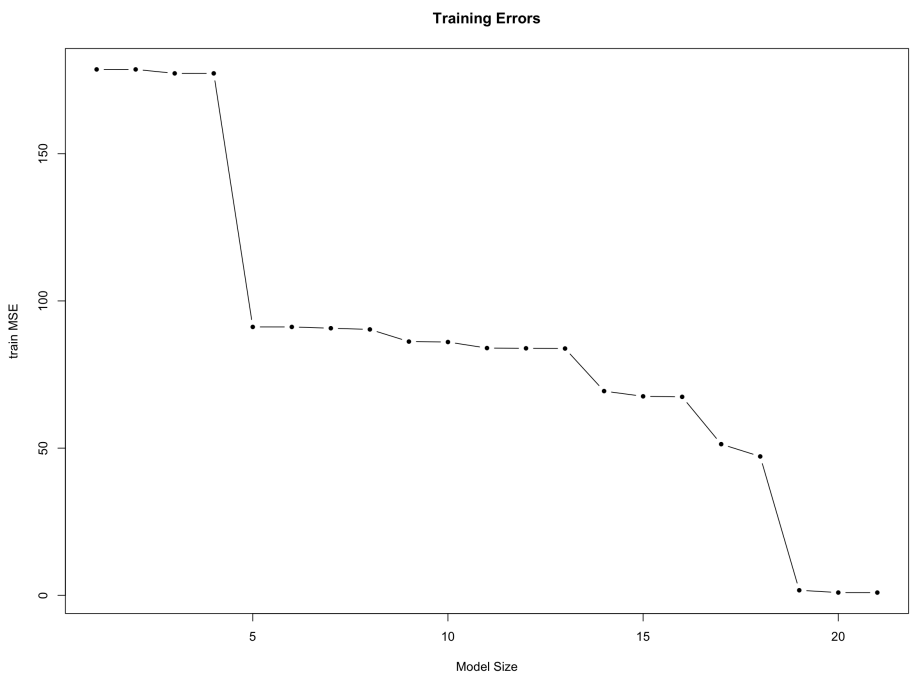
```
p = 20
n = 1000
error = rnorm(n,0,1)
beta = c(2, rep(0,5),1.2,3,-2,1.5,7,rep(0,5),3,4,1,9,2)
Xmat = matrix(rnorm(n*20),n,20)
X = cbind(rep(1,n),Xmat)
Y = X%*%beta + error
data = data.frame(Y,Xmat)
```

Note that `X%*%beta` is just doing matrix multiplication: `beta0 + beta1*X1 + beta2*X2 + .... beta20*X20`.

(a) Split your data set into a training set containing 100 observations and a test set containing 900 observations.

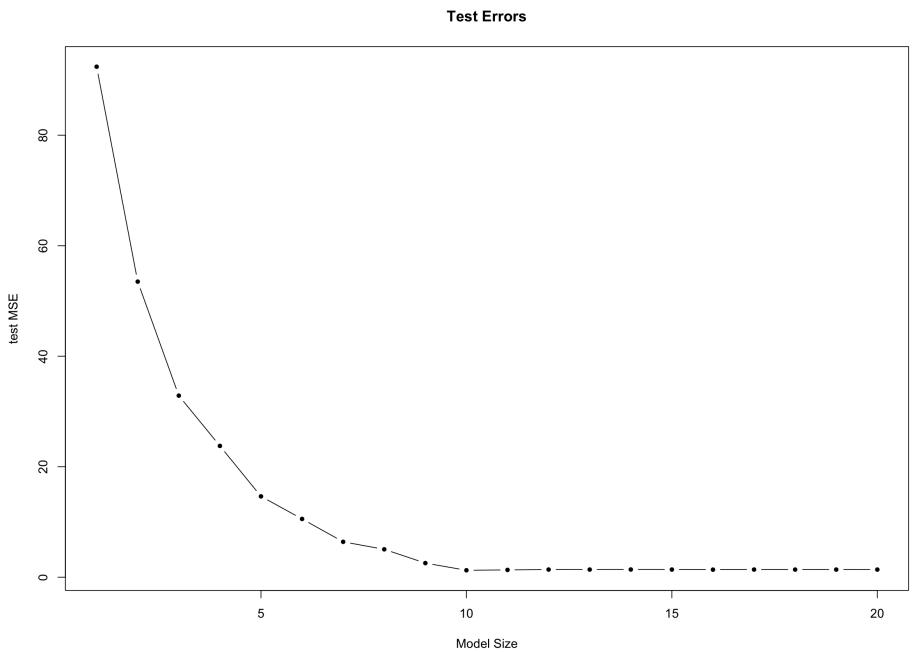
```
> train_index = sample(1:n,100,replace=FALSE)
> train = data[train_index,]
> test = data[-train_index,]
> dim(train)
[1] 100  21
> dim(test)
[1] 900  21
```

(b) Perform best subset selection on the training set only, and plot the training set MSE associated with the best model of each size.




```
best.fit = regsubsets(Y~.,data=train,nbest=1,nvmax=20)
plot(best.fit$rss/100, pch=20, main='Training Errors',type='b',
 ylab='train MSE', xlab='Model Size')
```

(c) Plot the test MSE associated with the best model of each size.




```
test.errors = rep(NA,20)
for(i in 1:20){
  test.mat = model.matrix(Y~.,data=test)

  coefi = coef(best.fit,id=i)
  pred = test.mat[,names(coefi)]%*%coefi
  test.errors[i] = mean((test$Y-pred)^2)
}

plot(test.errors,pch=20, main='Test Errors',type='b',
ylab='test MSE',xlab='Model Size')
```

(d) For which model size does the test MSE take on its minimum values? Comment on your results.

```
> which.min(test.errors)
[1] 10
> coef(best.fit,10)
(Intercept)          X6          X7          X8          X9         X10
  1.9843342   1.1099442   2.8258445  -1.9432324   1.3208752   7.0817983
        X16         X17         X18         X19         X20
  3.0266003   3.9332936   0.9142106   9.1416294   2.0028024
```

(e) How does the model at which the test MSE is minimized compare to the true model used to generate the data? Comment on how well the estimated regression coefficients match the true values.

Note: The true model contains only 10 predictors with nonzero coefficients. The remaining

10 predictors have true coefficient value equal to zero.

My test MSE was minimized for M_{10}, which perfectly recovered the true model I had generated in part (a). The regression coefficient values are very close, but not exact.

Problem 3: Forward and backward selection

We will use the `College` data set in the `ISLR2` library to predict the number of applications (`Apps`) each university received. Use all remaining variables as predictors.

(a) Randomly split the data set so that 90% of the data belong to the training set and the remaining 10% belong to the test set. Implement forward and backward selection on the training set only. Report the test MSE for all models $(M_1, M_2, \ldots, M_p)$ based on forward selection. Which model minimizes test MSE?

Report the test MSE for all models $(M_1, M_2, \ldots, M_p)$ based on backward selection. Which model minimizes test MSE?

Do both algorithms lead you to the same model based on test MSE?

Solutions may vary depending on how students split the data.

```
> library(ISLR2)
> n = dim(College)[1]
> train = sample(1:n,0.9*n,replace=FALSE)
> train_College = College[train,]
> test_College = College[-train,]
>
> library(leaps)
> regfit.fwd = regsubsets(Apps~.,data=train_College,nvmax=17, method="forward")
> regfit.bwd = regsubsets(Apps~.,data=train_College,nvmax=17, method="backward")
>
> test.MSE.fwd = rep(NA,17)
> for(i in 1:17){
+   test.mat = model.matrix(Apps~.,data=test_College)
+
+   coef.m = coef(regfit.fwd,id=i)
+
+   pred = test.mat[,names(coef.m)]%*%coef.m
+   test.MSE.fwd[i] = mean((test_College$Apps-pred)^2)
+ }
> test.MSE.fwd
 [1] 1438419 1122337 1128668 1092797 1042874 1048369 1097309 1067450 1056955 1022859
[11] 1006741 1017539 1009391 1010726 1014394 1012522 1012667
> which.min(test.MSE.fwd)
[1] 11
```

The model that minimizes test MSE based on forward selection is M_{11}:

```
coef(regfit.fwd,11)
 (Intercept)    PrivateYes        Accept        Enroll     Top10perc     Top25perc
-112.13551240 -492.94895762    1.58106425   -0.97406613   49.86592292  -16.05957331
 F.Undergrad      Outstate    Room.Board           PhD        Expend     Grad.Rate
   0.09240179   -0.09013116    0.14379565   -9.45504238    0.07522371    7.66764095

> test.MSE.bwd
 [1] 1960891 1444152 1279401 1172865 1215296 1227176 1235532 1228677 1206435 1240968
[11] 1198168 1214254 1206413 1216493 1214718 1214851 1214893
> which.min(test.MSE.bwd)
[1] 4
> coef(regfit.bwd,4)
 (Intercept)        Accept     Top10perc      Outstate        Expend
-693.39369139    1.44121435   31.33344783   -0.06676655    0.06408927
```

At least for my test/train split, backward and forward selection led to different models based on test MSE.

(b) Run forward and backward selection on the full dataset. For each approach, report the best model based on AIC. Do they lead you to the same model based on AIC?

They lead me to the same models. Forward selection based on AIC chose the following predictors:

```
> coef(regfit.fwd.full,12)
  (Intercept)    PrivateYes        Accept        Enroll     Top10perc     Top25perc
-157.28685883 -511.78760196    1.58691470   -0.88265385   50.41131660  -14.74735373
  F.Undergrad   P.Undergrad      Outstate    Room.Board           PhD        Expend
   0.05945481    0.04593068   -0.09017643    0.14776586  -10.70502848    0.07246655
    Grad.Rate
   8.63961002
```

Backward selection based on AIC also chose the same predictors:

```
> coef(regfit.bwd.full,12)
  (Intercept)    PrivateYes        Accept        Enroll     Top10perc     Top25perc
-157.28685883 -511.78760196    1.58691470   -0.88265385   50.41131660  -14.74735373
  F.Undergrad   P.Undergrad      Outstate    Room.Board           PhD        Expend
   0.05945481    0.04593068   -0.09017643    0.14776586  -10.70502848    0.07246655
    Grad.Rate
   8.63961002
```

(c) Which is the final model that you would pick? If both (a) and (b) lead you to the same model, then this is easy. Otherwise, there is no absolute right answer. Try to justify your choice using some ideas/concepts you've learned from this class.

Solutions can vary but they should be well-justified. For me, I prefer the model recovered using AIC. Backward and forward selection led me to the same model and I got to use all the data, so my results do not depend on the test/training split.

Problem 4: A Puzzling Problem

When fitting a linear regression model on a data set, you encounter the following `R` output. You notice there is something strange about the results. Point out what is strange in this output and **explain clearly** how this could happen.

```
Call:
lm(formula = y ~ x1 + x2)

Residuals:
    Min      1Q  Median      3Q     Max
-6.3700 -1.6364 -0.1208  1.4261  5.2558

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.2936     0.5217   4.396 2.82e-05 ***
x1            1.2600     2.3006   0.548    0.585
x2            1.8968     2.5509   0.744    0.459
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.376 on 97 degrees of freedom
Multiple R-squared:  0.09896, Adjusted R-squared:  0.08038
F-statistic: 5.326 on 2 and 97 DF,  p-value: 0.006385
```

We have a significant F-test, which indicates our model is an improvement over the null model, but the individual t-tests are not significant. At first this may seem contradictory but students should recognize this is due to multicollinearity: X_1 and X_2 are highly collinear, making it difficult to separate out their effects. More precisely, since their standard errors are quite large, they have reduced power to reject the null hypothesis. Therefore, neither individual predictor appears significant here.

Problem 5: Interaction Terms

We will use the `Credit` dataset for this problem. It is part of the library `ISLR2`.

(a) This data set contains a few categorical predictors. As we already discussed in lecture, these predictors should be stored as `factors` so that `R` can handle them properly. Using the `str` function, check that all the qualitative predictors in our dataset are stored correctly in `R` as factors. Copy and paste your output.

```
 > str(Credit)
'data.frame': 400 obs. of  11 variables:
 $ Income   : num  14.9 106 104.6 148.9 55.9 ...
 $ Limit    : num  3606 6645 7075 9504 4897 ...
 $ Rating   : num  283 483 514 681 357 569 259 512 266 491 ...
 $ Cards    : num  2 3 4 3 2 4 2 2 5 3 ...
```

```
$ Age      : num  34 82 71 36 68 77 37 87 66 41 ...
$ Education: num  11 15 11 11 16 10 12 9 13 19 ...
$ Own      : Factor w/ 2 levels "No","Yes": 1 2 1 2 1 1 2 1 2 2 ...
$ Student  : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1 1 1 1 2 ...
$ Married  : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 1 1 1 1 2 ...
$ Region   : Factor w/ 3 levels "East","South",..: 2 3 3 3 2 2 1 3 2 1 ...
$ Balance  : num  333 903 580 964 331 ...
```

(b) Fit a model with the response (Y) as credit card balance and X_1= `Income` and X_2 = `Student` as the predictors. Call this model `fit`. Summarize your output.

	Estimate	Standard Error	t-test	p-value
Intercept	211.1430	32.4572	6.505	2.34×10^{-10}
Income	5.9843	0.5566	10.751	$< 2 \times 10^{-16}$
StudentYes	382.6705	65.3108	5.859	9.78×10^{-09}

(c) Based on our results from part (b), write out the fitted model for students and write out the fitted model for non-students.

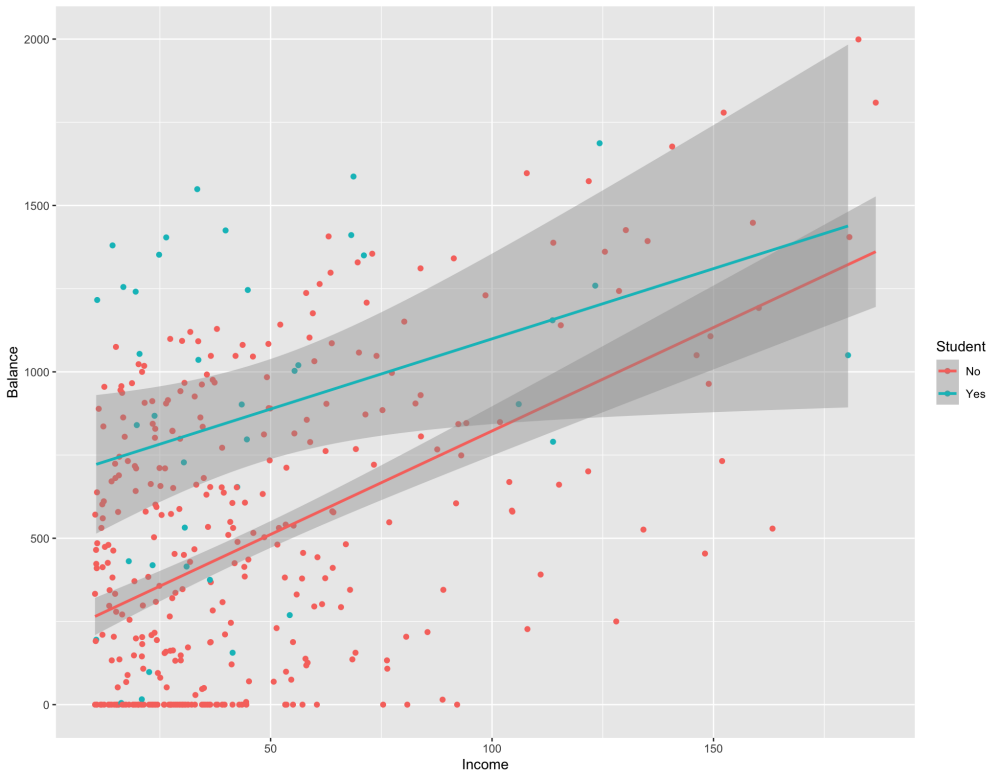
$$\hat{Y} = \begin{cases} 211.1430 + 5.9834 \times Income & \text{if individual is not a student,} \\ 593.7037 + 5.9834 \times Income & \text{if individual is a student.} \end{cases}$$

(d) Interpret the regression coefficient related to `Income` for both models.

A one unit increase in income is associated with 5.9834 ***average*** *increase in credit card balance, regardless of student status.*

(e) Notice that our model says that regardless of student status, the effect of `Income` on average `Balance` is the same. Do you think this is a reasonable constraint on our model? Construct some plots to back up your answer.

This is not a reasonable constraint. We would expect the relationship between income and credit card balance to be quite different for students versus non-students. From the plot below, it is clear that students versus non-students should be allowed to have different slopes between income and credit card balance (the lines are not parallel).



(f) One way we could relax this assumption is by incorporating *interaction terms* into our model. Specifically:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \epsilon_i,$$

where $X_1 =$ `Income`, $X_2 =$ `Student`, and $X_3 =$ `Income` $\times$ `Student`. Fit a model with an interaction term using the following code:

```
lm(Balance ~ Income + Student + Income:Student, data=Credit)
```

Based on this model, write out the fitted model for students and write out the fitted model for non-students.

$$\hat{Y} = \begin{cases} 200.62 + 6.22 \times \mathit{Income} & \text{if individual is not a student,} \\ 677.299 + 4.219 \times \mathit{Income} & \text{if individual is a student.} \end{cases}$$

(g) Interpret the regression coefficient related to `Income` for the fitted models obtained in part (f).

A one unit increase in income is associated with 6.22 ***average*** *increase in credit card balance for non-students while a 1 unit increase in income is associated with 4.219* ***average*** *increase in credit card balance for students.*

End of solutions.